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Critique of Pure Reason

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Pokémon forti, Pokémon deboli.
Sono distinzioni dettate dall'egoismo.
Gli allenatori davvero in gamba dovrebbero vincere coi loro preferiti.

Strong Pokémon, weak Pokémon.
That is only the selfish perception of people.
Truly skilled trainers should try to win with their favorites.

Karen, Johto Elite Four

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Your introduction text here.

Chapter 1

Theoretical Background

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1.1 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [1], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.1.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.1.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c.\ , \tag{1.1}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.2}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.3}$$

Averaging over a field period [1]:

$$U_{\text{dip}} = -\frac{1}{2}\langle\mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha]I, \tag{1.4}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.
It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}} = \frac{1}{2\epsilon_0 c}\text{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.5}$$

The dipole oscillation absorbs a power

$$P_{\text{abs}} = \langle\dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega\text{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\text{Im}[\alpha]I, \tag{1.6}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{P_{\mathrm{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\mathrm{Im}[\alpha]I(\mathbf{r}). \tag{1.7}$$

1.1.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.8}$$

Using $E(t) = \mathcal{E}_0 \exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0 \exp(-i\omega t)$, equation (1.8) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.9}$$

Using $p = -ex(t) = -ex_0 \exp(-i\omega t)$ and (1.9), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.10}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.11}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\,\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.12}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}|\langle e|\,\hat{d}\,|g\rangle|^2, \tag{1.13}$$

where $\hat{d}$ is the dipole moment operator.

Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.8 (for a rigorous derivation see Chapter V of Steck [2]). Therefore, the classical polarizability given by equation 1.12 represent a well suited approximation of ODTs.

1.1.1.3 Dipole potential and scattering rate

Substituting α form (1.12) in (1.4) and (1.7) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right)I(\mathbf{r}), \tag{1.14}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.15}$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of ~ 3.8 THz thus a trap typically works at few Thz of detuning, while $\Gamma \sim 36$ MHz. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I(\mathbf{r}) \tag{1.16}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\text{dip}} \tag{1.17}$$

Equations (1.16) and (1.17) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.1.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.1.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}} \cdot \mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j \neq i} \frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.18}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + \hbar(n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.18) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.19}$$

where Δ_{ij} is the detuning of the $i \rightarrow j$ transition with respect to the laser field.

Considering a two-level atom first, (1.18) has only one term, using equations (1.13) and (1.19) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm\frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm\frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.20}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.20) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.16). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. If a low saturation regime is assumed ($I/\Delta \ll 1$), all the atom can be considered in the ground state, and the light-shifted ground state became the effective potential that govern the motion of the atoms which drives them toward region of minima of ΔE (Figure 1.1).

For a multi-level atom equation (1.18) contains many terms similar to equation (1.20), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. However, each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\hat{d}|e_j\rangle$. Such quantity can be expressed as

$$d_{ij} = c_{ij}||d|| \tag{1.21}$$

where $||d||$ is the reduced matrix element given by (1.13) and c_{ij} are real transition coef-

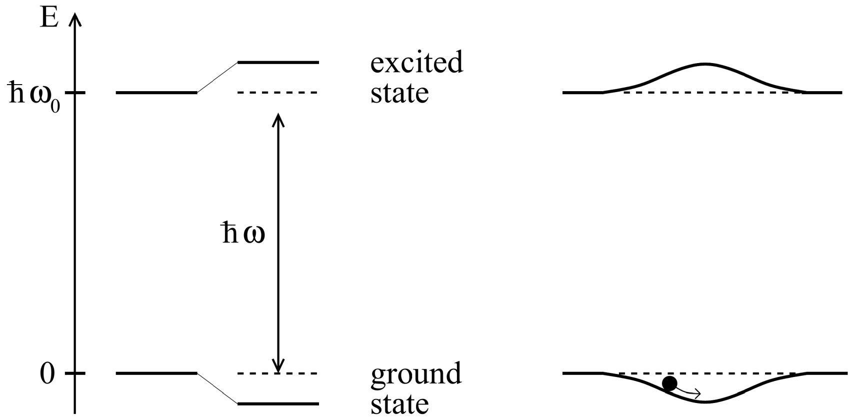

Figure 1.1: *From [1]. Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms.*

ficients that take into account the coupling strength of the two involved states $|g_i\rangle$ and $|e_j\rangle$, and depends on their angular momentum value [1][3].
Therefore, the energy shift (1.18) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.22}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.23}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transition, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.1.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the VIS-IR range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structure are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.
When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is $^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

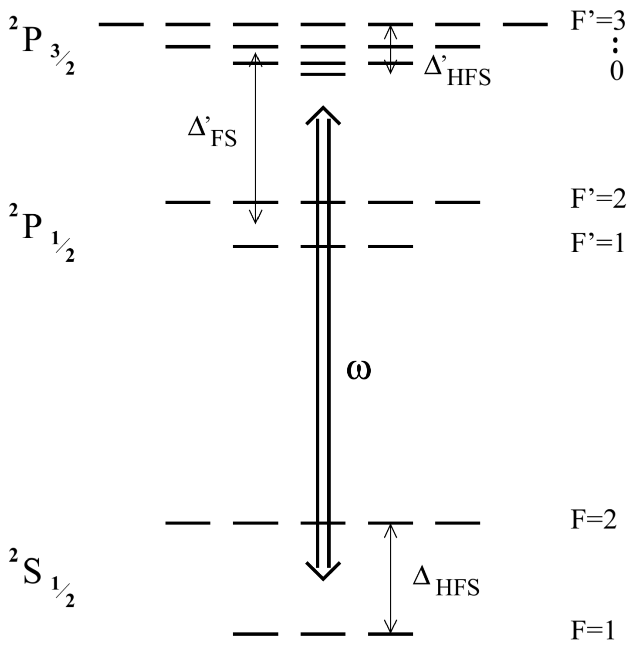

Figure 1.2: *From [1]. Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb.*

- D_1: $^2S_{1/2} \rightarrow {}^2P_{1/2}$;
- D_2: $^2S_{1/2} \rightarrow {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.2.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.22) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [1]:

$$U_{\mathrm{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P} g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P} g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.24}$$

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^{\pm}$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg$

Δ'_{FS}), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.24) at first order of the small parameter Δ'_{FS}/Δ, one gets [1]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}\left(1 + \frac{1}{3}\mathcal{P}g_F m_F \frac{\Delta'_{\text{FS}}}{\Delta}\right) I(\mathbf{r}). \tag{1.25}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.16)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.25) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [1]. Therefore, linearly polarized light is usually employed for dipole trap implementation.

1.1.3 Heating mechanism

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.1.3.1 Heating source

When atoms interact with laser light, fluctuations in the spontaneous emission and absorption always causes heating (this is also the phenomenon that puts a lower limit to cooling mechanism [4][3]). For the optical dipole trap the atoms absorbs photons and re-emits them in random directions (isotropic radiation), statistical oscillations in both these processes cause heating. The absorption of photons from the dipole light is associ-

ated with a recoil energy $E_{\text{rec}} = k_B T_{\text{rec}}/2$ per scattering event and happens only in the direction of the laser beam, hence it is anisotropic. On the other hand, also the re-emition scattering events cause a recoil energy E_{rec}, however the direction of emission is random, thus the effect is spread over all the three spatial direction. Thus, for a single cycle the atom receives E_{rec} from the absorption and then E_{rec} from the spontaneous emission, for a total of E_{rec}, in a time $\bar{\Gamma}_{\text{sc}}^{-1}$.
It leads to the total heating power:

$$P_{\text{heat}} = 2E_{\text{rec}}\bar{\Gamma}_{\text{sc}} = k_B T_{\text{rec}}\bar{\Gamma}_{\text{sc}} \tag{1.26}$$

thus the heating depends on the mean scattering rate $\bar{\Gamma}_{\text{sc}}$ and on the recoil E_{rec} which depends on the laser frequency.
Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [1].

1.1.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.27}$$

Using equation (1.26) in equation (1.27), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3}T_{\text{rec}}\bar{\Gamma}_{\text{sc}}. \tag{1.28}$$

Expressing the dipole potential as

$$\bar{U}_{\text{dip}} = U_0 + \bar{E}_{\text{pot}} = U_0 + \frac{3}{2}k_B T, \tag{1.29}$$

where U_0 is the potential energy minimum, then the scattering rate $\bar{\Gamma}_{\text{sc}}$ can be expressed by Substituting U_{dip} from (1.17), obtaining

$$\bar{\Gamma}_{\text{sc}} = \frac{\Gamma}{\hbar\Delta}(U_0 + \frac{3}{2}k_B T). \tag{1.30}$$

It is important to notice that for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, while for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.30) in (1.28), with a depth $\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\mathrm{red}} = \frac{1}{3} T_{\mathrm{rec}} \frac{\Gamma}{\hbar|\Delta|} \hat{U}, \tag{1.31a}$$

$$\dot{T}_{\mathrm{blue}} = \frac{1}{2} T_{\mathrm{rec}} \frac{\Gamma}{\hbar\Delta} k_B T. \tag{1.31b}$$

Equation (1.31a) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

1.2 Optical Fibers

1.2.1 Circularly symmetric wave guides

1.2.2 Weak waveguide approximation

1.2.3 The LP modes

Chapter 2

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present thesis work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT.
Although beyond the scope of the current work, it is worth noting that the long-term objective of this experiment is to confine the atoms within the core of a hollow-core fiber.

2.1 Magneto-optical trap

2.1.1 The Vacuum Chamber

2.1.2 The laser system

2.1.2.1 Master laser

2.1.2.2 Cooler and repumper lasers

2.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 2.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.
During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following subsections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

2.2.1 Optical setup

In this experiment the ODT light is guided through a HCPCF, which introduces a critical complication in the injection procedure. Indeed, in contrast with standard telecom fibers, HCPCFs are multi-mode fibers. As a consequence, even small deviations of the injected beam from the ideal injection angle excites higher order modes in addition to the fundamental one. This results in distorted output beam profiles, which are not well suited for the realization of an optical conveyor belt based on the superposition of the beam guided by the HCPCF and a beam guided through a conventional telecom fiber supporting only the fundamental LP_{01} mode.
For this reason, particular cure was devoted to the alignment procedure. A significant improvement in the quality of the transmitted mode was achieved by optimizing the injection while inspecting the near-field of the transmitted beam, rather than simply maximizing the transmission efficiency. The following sections describe the injection setup and the horizontal microscope built for the near-field inspection.

2.2.1.1 Injection setup

The ODT is realized by employing the laser IPG Photonics YLR-20-1064-LP-SF, which is an Yitterbium fiber laser that emits 1064 nm linearly polarized light at about 3 W of power.
Figure 2.1 shows the optical setup scheme used for the injection of the laser light in the HCPCF. The scheme is composed of the following elements:

- **Half Wave Plate 1** (HWP1) and **Optical Isolator**. The optical isolator prevents back reflections from propagating towards the laser cavity. High-power lasers are particularly sensitive to back reflections, which can introduce power instabilities and

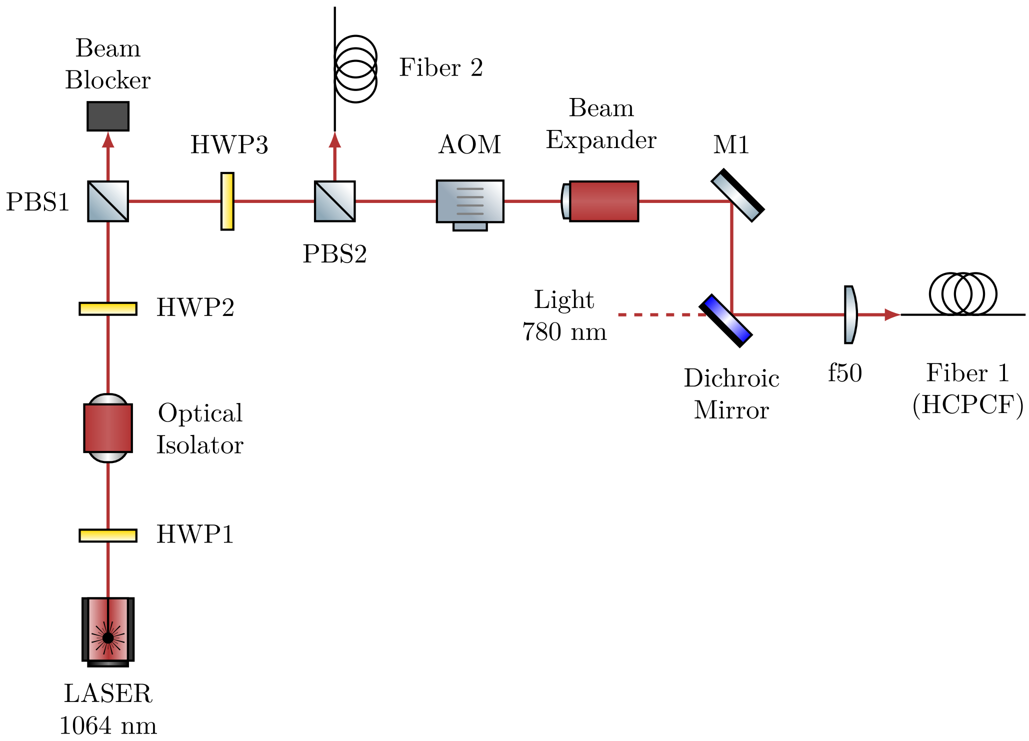

Figure 2.1: *Scheme of the optical setup for 1064 nm ODT beam injection into the HCPCF. The Acousto-Optic Modulator (AOM) operates at 110 MHz, and the subsequent setup utilizes the fist diffraction order. The order zero is wasted on a beam blocker (not drawn). The 780 nm light setup is not drawn, its presence will be important in the future stages of the experiment for the characterization of the atoms inside the HCPCF core. Fiber 2 injection setup is not drawn as well.*

potentially damage the laser system. Since optical isolators operate correctly only for a specific input polarization, HWP1 is used to optimize the polarization of the incoming beam and maximize the transmission through the isolator.

- **HWP2** and **Polarization Beam Splitter 1** (PBS1). These components are used to regulate the optical power delivered to the subsequent stages of the setup. The unwanted fraction of the beam is directed onto a beam block. This configuration is particularly useful during alignment procedures, where the optical power is reduced to a few milliwatts for safety reasons.

- **HWP3** and **PBS2**. PBS2 splits the beam into two different branches. The first is employed for the HCPCF injection, while the second will be injected into a PANDA polarization-maintaining fiber, which eventually will provide the second beam entering the chamber from above for the realization of the optical conveyor belt. HWP3 is adjusted to have approximately 50/50 power splitting ratio, although this value can be modified if required in future stages of the experiment.

- **Acousto-Optic Modulator** (AOM). The AOM is employed as a fast switch for the ODT beam. The device operates using an acoustic wave at 110 MHz, causing the first diffraction order to be frequency shifted by the same amount. Although this shift is negligible with respect to the trap depth of the ODT, it will become relevant in the future implementation of the optical conveyor belt.

- **Beam expander**. A ∼3X beam expander is employed to increase the beam diameter and collimate it before the focusing conditions for coupling into the fiber.

- **Mirror 1** (M1) and **Dichroic Mirror**. Both mirrors are mounted on tilt stages and their angular adjustment is used for optimizing the fiber injection alignment. The dichroic mirror also combines the 1064 nm ODT beam with a 780 nm beam that will be used in the future for the characterization of the atoms trapped inside the HCPCF core.

- **Focusing Lens** (f50). A lens with focal length $f = 50$ mm focuses the beam into the HCPCF. It is mounted on a horizontal translation stage, providing an additional degree of freedom for optimizing the injection alignment.

- **Fiber 1** (HCPCF). The HCPCF is mounted on a x-y translation stage and terminated with an FC/APC connector. The translation stage allows transverse displacement of the fiber, providing additional alignment degrees of freedom during the injection procedure, although these degrees of freedom were found to have limited impact on the final optimization.

TO BE CONTINUED, here add the calculated mode radius at fiber injection, using results from the laser section

2.2.1.2 Horizontal microscope

Initially, the injection into the HCPCF installed inside the vacuum chamber was attempted by using a Complementary Metal-Oxide-Semiconductor (CMOS) camera to monitoring the beam profile. However, despite the effort, strongly deformed mode profiles were obtained.

In order to better investigate the injection procedure and simplify the alignment process, a separate test HCPCF was employed outside the vacuum chamber. The test fiber consisted of an approximately 50 cm HCPCF segment with the same FC/APC connector as the fiber in the chamber. The test fiber allowed an easy measurement of the efficiency on the exit facet of the beam without any window in the path, and being outside the void chamber allow for easier mode imaging. Moreover, since the two fibers shared the same connector, the optimal alignment conditions identified with the test fiber were expected to provide a good starting point for the alignment of the HCPCF installed inside the chamber. Nevertheless, even with the test fiber, optimization based solely on transmission

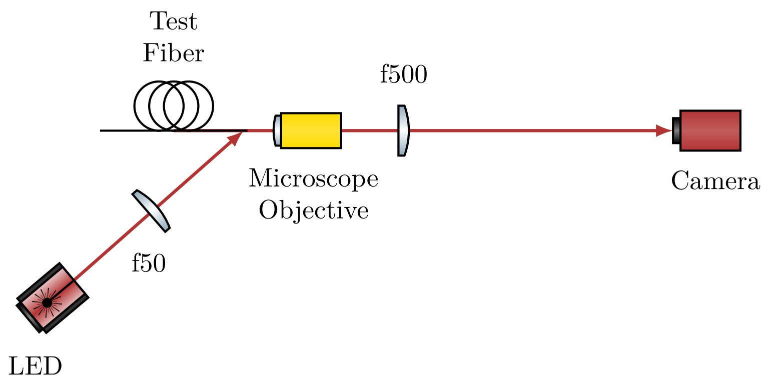

Figure 2.2: *Scheme of the optical setup for the horizontal microscope.*

efficiency and far-field inspection still resulted in poor beam quality and low efficiency of $\sim 30\%$.

The fiber manufacturer suggested that the optimal injection condition should be determined by inspecting the shape of the near-field of the transmitted mode. For this purpose, a horizontal microscope system was developed to inspect the HCPCF facet and to investigate the near-field profile of the transmitted mode. The optical scheme of the microscope is shown in Figure 2.2.

The fiber was positioned inside a 90° V-groove of depth 350 μm. The fiber was fixed inside the groove by means of a thin foam layer attached to a tiltable mount using adhesive tape. The coating was removed from the fiber end, leaving approximately 15 mm of suspended fiber in order to allow lateral illumination of the facet.

The HCPCF facet was illuminated by a red LED focused by an $f = 50$ mm lens onto the final suspended portion of the fiber. Part of the red light propagated through the silica structure forming the kagome lattice, making them visible on the microscope camera. The observation of the kagome structure was used to optimize the microscope focus and to inspect the integrity and cleanliness of the fiber facet.

The light emerging from the fiber was collected by a optical microscope objective lens (Olympus PLN10X objective, nominal focal length $f = 18$ mm, working distance $d =$ 10.6 mm) and subsequently re-imaged onto a CMOS camera by means of an $f = 500$ mm lens. The expected magnification is $M = f_{\text{tube}}/f_{\text{lens}} \approx 27.8$. The camera was connected to a computer for real-time imaging of the transmitted near-field mode. In addition, a powermeter could be inserted between the fiber output and the objective lens in order to evaluate the coupling efficiency by comparing the transmitted power with the optical power measured before the injection stage. Finally, a black paper tube was placed between the imaging $f = 500$ mm lens and the camera to reduce background light from the LED illumination and from ambient room light.

The alignment procedure was carried out by optimizing the transmitted near-field profile in order to obtain a mode as close as possible to a single-lobed structure, results and further details are found in Section 2.2.3.

Figure 2.3 shows the top-down photographs of the complete experimental apparatus, including both the injection setup and the horizontal microscope system, and figure 2.4 shows a detailed view of the horizontal microscope.

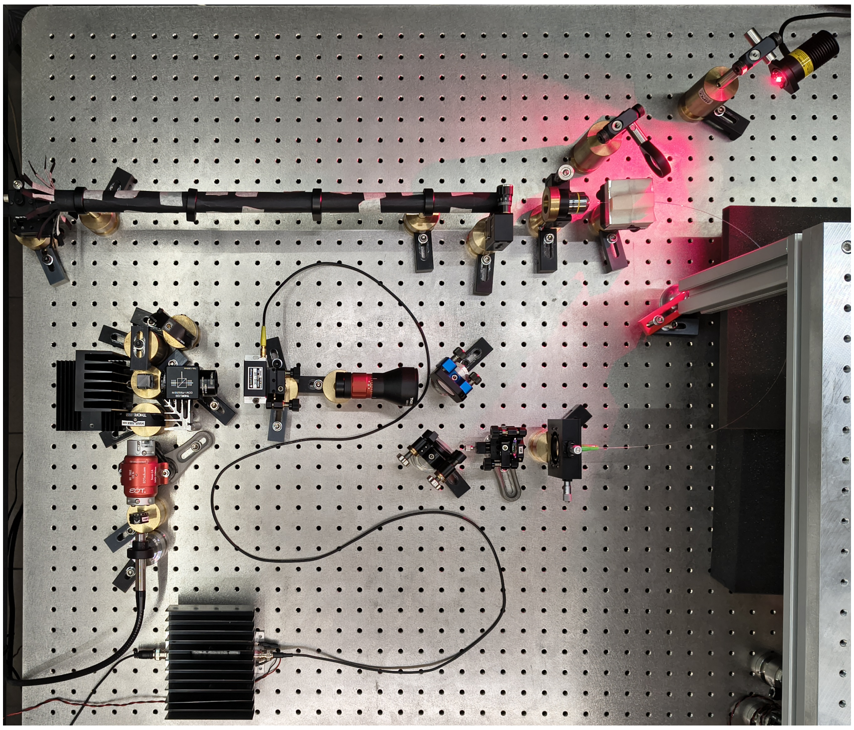

Figure 2.3: *Top-down photograph of the complete experimental apparatus. The horizontal microscope, shielded by the black paper tube, is visible at the top of the image. The optical injection setup is located in the center, while the RF amplifier driving the AOM is placed at the bottom. For reference, the distance between the screw holes on the breadboard is 2.5 cm.*

2.2.2 1064nm laser characterization

The ODT is realized by employing the laser IPG Photonics YLR-20-1064-LP-SF, which is a Ytterbium fiber laser that emits 1064 nm linearly polarized light. The mode field diameter of the laser changes significantly upon changes of the nominal power set, therefore the mode has been characterized as a function of the power set to allow a correct realization of the HCPCF injection setup. The calibration will also provide a reference for future fine tunings of the HCPCF injection.

TO BE CONTINUED, it will be moved before the optical setup

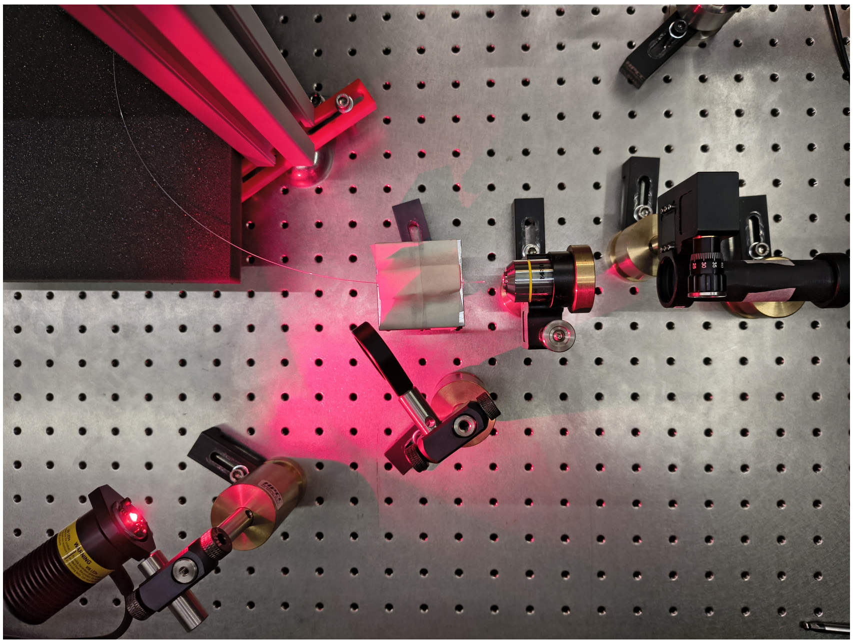

Figure 2.4: *Detailed top-down view of the horizontal microscope. The setup highlights the test fiber, the red LED providing lateral illumination to the fiber facet, the microscope objective lens and the refocusing lens.*

2.2.3 Injection alignment and near-field imaging

For a conventional single mode telecom fiber, the alignment procedure is straight forward: it is sufficient to inject a low power beam of few milliwatts, monitoring the power at the exit of the fiber and adjust the injected beam maximizing the guided optical power.

By contrast, for a multimodal HCPCFs the alignment procedure is not straight forward. The silica jacket that surrounds the kagome core can guide the light with good coupling efficiencies of 60%, although with badly shaped mode (see figure 2.5). Given also the large dimension of the jacket compared to the core, monitoring just the coupling efficiency is particularly ineffective, because one can get stacked in the local maxima efficiency of the jacket.
As suggested by the manufacturer, the best procedure for the injection alignment resulted to be monitoring the near-field mode at the fiber exit. The full alignment procedure consists into use the powermeter first to obtain a signal which guarantees that the impinging beam is on the fiber, then switch to the live monitoring of the near-field and optimize its shape by regulating the degrees of freedom of the injection setup. The coupling efficiency is then measured only at the end of the mode shape optimization procedure.

Using the procedure described above, the test fiber injection has been optimized, employing the injection optical setup and the horizontal microscope described in Section 2.2.1. This led to a good result and thus the same procedure has been applied to align the fiber inside the vacuum chamber.

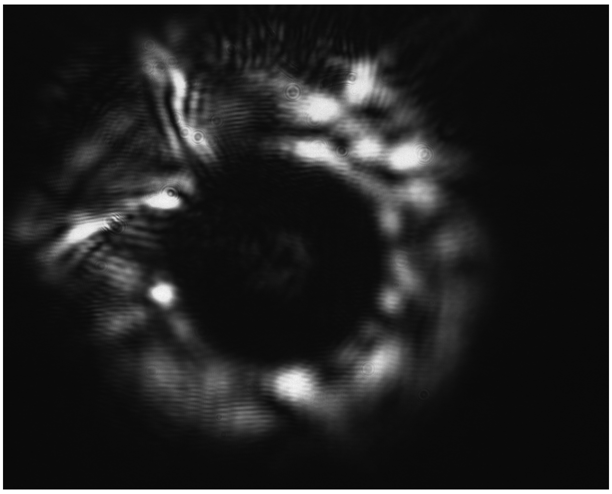

Figure 2.5: *Example of the near-field of a mode guided by the test fiber jacket. The light is mainly guided by the silica jacket surrounding the kagome structure, while just a weak signal is visible in the core. Note that this strongly deformed mode had a coupling efficiency of* $\sim 60\%$.

2.2.3.1 Test fiber

2.2.3.2 Chamber fiber

2.2.4 MOT to ODT loading sequence

Chapter 3

Result And Discussion

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